

Main loop

Your turtles are now set up and ready to go. Let's write the code that makes the game come to life.



EXPERT TIPS

Pass

In Python, if you're not yet sure what code you want inside a function, you can just type in the **pass** keyword and then come back to it later. It's a bit like passing on a question in a quiz.

6 Placeholder functions

You can put off defining a function until later by using the **pass** keyword. Under the code for the turtles, add the following placeholders for functions that you'll fill with code in later steps.

```
def outside_window():
    pass

def game_over():
    pass

def display_score(current_score):
    pass

def place_leaf():
    pass
```

To get a basic version of the program running sooner, you can use placeholders for functions that you'll finish coding later.

7 Game starter

After the four placeholder functions comes the **start_game()** function, which sets up some variables and prepares the screen before the main animation loop begins. You'll add the code for the main loop, which forms the rest of this function, in the next step.

The turtle stretches into a caterpillar shape.



```
def start_game():
    global game_started
    if game_started:
        return
    game_started = True

    score = 0
    text_turtle.clear()

    caterpillar_speed = 2
    caterpillar_length = 3
    caterpillar.shapesize(1, caterpillar_length, 1)
    caterpillar.showturtle()
    display_score(score)
    place_leaf()
```

If the game has already started, the return command makes the function quit so it doesn't run a second time.

Clear the text from the screen.

This line places the first leaf on the screen.